Bilal Sabugar

Robotics | Mechatronics | Control & Software

bilal.07.04.07@gmail.com | Portfolio | GitHub | LinkedIn

B.Tech student working across mechanical design, robotics platforms and engineering software. Co-founder and CTO of Aetheron Labs, with research internship experience at the IITGN Robotics Lab.

EXPERIENCE

Research internship | IITGN Robotics Lab, IIT Gandhinagar

Worked with Aditya Jadav and Archie Veera on a tactile-based ball-balancing system using a half-humanoid platform. Related OSRS responsibilities include platform engineering, online system identification and full-stack robotics.

Co-founder & CTO | Aetheron Labs

Mechanical design, structural fabrication and mechatronic integration within the founding team. Work connects the physical structure, actuators and electronics across company robotics programs.

SELECTED ENGINEERING WORK

Novexa | 18-joint hexapod

Contributed to mechanical design, fabrication and integration within a four-person team. Evidence includes original Fusion 360 CAD, printable parts, a physical prototype and a co-authored journal paper. Case study

OSRS | Robotics research platform

Credited for platform engineering, RLS identification and full-stack robotics. The shared platform connects robot hardware, perception, MuJoCo simulation and telemetry. Source and contribution

RoboScout | Engineering research software

Directed the product and engineering workflow for source-linked component research and deterministic sizing calculations. Implementation uses AI coding agents under human architectural direction. Architecture

Additional software: WIRK (task and focus workspace), RecoverOS (test-mode payment recovery), Calm Pulse (team wellness prototype; back-end development).

EDUCATION

ATLAS SkillTech University | B.Tech, 2025-2029 (expected)

Computer Science, Artificial Intelligence and Robotics.

TECHNICAL TOOLKIT

Mechanical: Fusion 360, 3D printing, actuation and integration.
Robotics: Python, OpenCV, MuJoCo, control interfaces and system identification.
Software: TypeScript, React, Next.js, FastAPI, React Native and Git.

PUBLICATION

Fabrication of a 3D Printed Hexapod for All Terrain Adaptive Locomotion. Co-authored with Aditya Jadav, Archie Veera and Rabinder Henry. IJIRT, Vol. 12, Issue 11, pp. 12393-12396, April 2026. Paper